

Problem statement: Ocean wind field estimation using SAR imagery over Indian coastal areas for wind farm planning

Study area: coast of Tamil Nadu or Gujarat

Deliverable: API based working code to provide wind field vectors over a given coastal region are of interest. User provides date and area of interest, the API provides the corresponding wind field vectors.

Team members: upto 2

Timeline: interim results and final results-08 June.

Dataset(source and used): Sentinel-1

Reference : 1. Wind direction retrieval from Sentinel-1 SAR images using ResNet Wind direction retrieval from Sentinel-1 SAR images using ResNet <https://www.sciencedirect.com/science/article/pii/S0034425720305514>; 2. Wind Field Estimation <https://step.esa.int/main/wp-content/help/versions/13.0.0/snap-toolboxes/eu.esa.microwavetbx.sar.op...>. Feel free to explore more advanced DL techniques as per your expertise and interest.

Python/GEE code repository: to be developed by student

Validation results table: to be developed by student

Final report: to be developed by student

Sample results:

